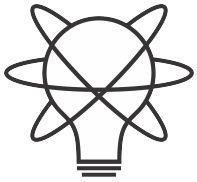
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Quantum pure states are not enough

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Category: Quantum mechanics - Tags: Quantum statistical mechanics, Convex spaces

Abstract

We show that the space of quantum pure states is not enough to uniquely recover the space of quantum ensembles, and therefore it is not enough to characterize quantum mechanics. The space of statistical ensembles, on the other hand, is enough to distinguish between the two.

1 Introduction

While most efforts within the foundations of quantum mechanics focus on pure states, we show they are not enough to characterize quantum mechanics. It is the space of statistical ensembles that marks the difference and, therefore, statistical mechanics plays a crucial role in the difference between the two theories.

2 Conditions

Since we want to compare the classical and quantum state space structures layer by layer, we break them up along similar lines: states and the convex structure of statistical ensembles.

The first two conditions just define the states, the full descriptions of the respective systems.

Condition CP (Classical state space only)**.** *The state space of a classical system is represented by a symplectic manifold* (X, ω)*.*

Condition QP-RAY (Quantum state space only)**.** *The state space of a quantum system is represented by the projective space* $\mathcal{P}(\mathcal{H})$ *of a complex Hilbert space* $\mathcal{H}$*.*

The next conditions specify the ensemble spaces and identify convex combinations with statistical mixtures.

Condition CE (Classical ensemble space)**.** *The statistical ensembles of a classical system are represented by probability measures over its state space. Statistical mixtures are represented by convex combinations of these measures.*

Condition QE-DENS (Quantum ensemble space)**.** *The statistical ensembles of a quantum system are represented by density operators* $\mathcal{D}(\mathcal{H})$*. Statistical mixtures are represented by convex combinations of density operators.*

Finally, we state the exclusivity condition whose relationship to these layers we want to investigate.

Condition CQE (Classical–quantum exclusivity)**.** *A system cannot be both classical and quantum.*

By system here we mean the model, not the physical object. For example, an electron could be modeled as either a classical or quantum particle depending on the circumstances. Conversely, a classical harmonic oscillator could be realized by masses on springs or electrical circuits. A system is assumed to have at least two distinct states, as a system with only one possible state cannot produce different outputs (e.g. cannot be present or not present) and, therefore, is indistinguishable as a separate system.

3 Theorems

In general, we want to understand whether CQE is guaranteed by the representations provided by the background conditions. The first theorem shows that, if we limit the mathematical representation to the state space definition, systems can be represented both by quantum Hilbert spaces and classical symplectic spaces.

Theorem 1. *Over CP+QP-RAY, CQE is not guaranteed.*

Proof. Recall that finite-dimensional complex projective spaces are symplectic spaces when equipped with the Fubini–Study symplectic form. Consider, then, the state space for a fixed-magnitude magnetic dipole, described by its orientation. Under CP, this will be represented by a two-sphere, which is both a symplectic manifold and a complex projective line. Similarly, consider the state space of a qubit. Under QP-RAY, this will be represented by a Bloch sphere, which is both a complex projective line and a symplectic manifold. Both these systems are non-trivial. Moreover, over CP+QP-RAY, both systems can be represented with both a classical and a quantum representation. This means that, over CP+QP-RAY, CQE is not guaranteed. □

The second theorem shows that, if we include ensembles, the mathematical representations of classical and quantum systems diverge.

Theorem 2. *Over CP+CE+QP-RAY+QE-DENS, CQE is guaranteed.*

Proof. For a system S to violate CQE over CP+CE+QP-RAY+QE-DENS, its state space X must first admit representations as both a symplectic manifold and a complex projective space. Suppose S has such a property. Let $|\psi_0\rangle, |\psi_1\rangle \in \mathcal{H}$ be two normalized orthogonal state vectors in terms of the quantum representation, and $|\phi_\pm\rangle = \sqrt{\frac{1}{2}}(|\psi_0\rangle \pm |\psi_1\rangle)$. Consider the equal mixtures of the respective pairs in terms of density operators. We have $\frac{1}{2}\rho_{\psi_0} + \frac{1}{2}\rho_{\psi_1} = \frac{1}{2}\rho_{\phi_+} + \frac{1}{2}\rho_{\phi_-}$. Now consider the equal mixtures of the respective pairs in terms of probability measures. We have $\frac{1}{2}\delta_{\psi_0} + \frac{1}{2}\delta_{\psi_1} \neq \frac{1}{2}\delta_{\phi_+} + \frac{1}{2}\delta_{\phi_-}$. That is, density operators allow multiple decompositions in terms of pure states while classical probability distributions do not. Given that, physically, a statistical mixture has a well-defined result, it cannot be represented in two inconsistent ways. Therefore, over CP+CE+QP-RAY+QE-DENS, CQE is guaranteed. □

Note that the ensemble space structure of quantum mechanics does not automatically imply the Born rule, which is an additional condition. However, imposing the Born rule implies the structure of the ensembles. In fact, in textbooks, it is often constructed once the Born rule is imposed. Also,

while the Born rule is typically used to define measurements and unitary evolution, one can invert this as well. If one is given the space of all valid measurements and valid unitaries, one can recover the Born rule. In this sense, we have the following insight.

Insight 3. *The space of statistical ensembles is necessary to distinguish between classical and quantum spaces.*

Lastly, note that each quantum ensemble can be understood as an equivalence class of classical distributions over the pure states that give that particular ensemble. While the classical description allows any random variable, we can imagine fewer random variables are available. For example, if we take the case of a classical magnetic dipole and suppose that we can characterize states through the average magnetic moment, the space of ensembles would reduce to the Bloch ball.

4 Conclusion

Quantum mechanics cannot be understood by simply looking at pure states, their properties and their evolution. On the other hand, since the difference in the convex space of ensembles is enough to distinguish between the two theories, understanding the difference between classical and quantum mechanics requires understanding the difference between their statistical counterparts.